

Artificial Intelligence–Based Prediction of Prognosis After Nonsurgical Treatment for Medial Meniscus Posterior Root Tear

Hong Yeol Yang,* MD, PhD , Inhwon Bae,[†] PhD , Ji Won Kim,* MD ,
Gyunghwan Bae,* MD, Youzhen Zheng,* MD, Sung Ju Kang,* MD,
and Jong Keun Seon,*[‡] MD, PhD 

Investigation performed at Chonnam National University Hwasun Hospital, Hwasun, Republic of Korea

Background: Medial meniscus posterior root tear (MMPRT) is recognized as one of the leading causes of knee osteoarthritis. Given the detrimental effects of MMPRT on knee kinematics and the associated clinical consequences, substantial efforts have been directed toward improving the understanding and management of MMPRT.

Purpose: To develop and validate an artificial intelligence (AI)–based prediction model for patient-specific risk assessment of clinical failure at 2 and 5 years after nonsurgical treatment of MMPRT.

Study Design: Case-control study; Level of evidence, 3.

Methods: The authors retrospectively reviewed a prospectively collected database of 233 patients who underwent nonsurgical treatment for MMPRT between 2006 and 2020. Patient descriptive characteristics, clinical data, and imaging variables were evaluated for their association with clinical failure, defined as conversion to total knee arthroplasty or corrective osteotomy at 2- and 5-year follow-up. Five conventional machine learning models, including Elastic Net logistic regression, multilayer perceptron, support vector machine, random forest, and Extreme Gradient Boosting, as well as a proposed deep learning model, the Grouped Graph Attention (GGAT) network, were developed and internally validated to predict clinical failure.

Results: During follow-up, clinical failure occurred in 36 of 233 patients (15.5%) at 2 years and in 54 of 233 patients (23.2%) at 5 years. The deep learning–based GGAT model demonstrated better overall predictive performance compared with conventional machine learning models at both the 2- and 5-year follow-up, with test set accuracy of 0.83 to 0.91, precision of 0.87 to 0.91, sensitivity of 0.92 to 1.00, F1 score of 0.89 to 0.95, Brier score of 0.09 to 0.16, and areas under the receiver operating characteristic curve of 0.69 to 0.77. The most influential predictors of clinical failure included baseline mechanical hip-knee-ankle angle, symptom duration, body mass index, age, bone marrow edema, lateral distal femoral angle, subchondral insufficiency fracture of the knee, cartilage lesion, effusion grade, and medial proximal tibial angle.

Conclusion: The deep learning–based GGAT network demonstrated accurate prediction of clinical failure at both 2 and 5 years after nonsurgical treatment of MMPRT. These findings underscore the potential value of deep learning–based risk assessment in supporting clinical decision-making and patient counseling.

Keywords: medial meniscus posterior root tear; nonsurgical treatment; artificial intelligence; deep learning; prognostic modeling; clinical failure

Medial meniscus posterior root tear (MMPRT) is defined as an avulsion injury or radial tear involving the posterior bony attachment.⁵¹ It has been shown to be biomechanically equivalent to complete meniscectomy, thereby

serving as a precursor to the rapid development of knee osteoarthritis.³⁷ Owing to the progressive nature of osteoarthritis and its deleterious clinical course, an increasing number of treatment strategies for MMPRT have been described, including meniscectomy, root repair, and non-surgical management.^{15,23,42}

Although meniscectomy has historically been a commonly used treatment for MMPRT, this approach does

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not restore meniscal hoop strain and leads to altered knee biomechanics with accelerated degeneration of the articular cartilage.^{15,16} In contrast, meniscus root repair has been shown to reestablish tibiofemoral contact mechanics to a state comparable to that of the intact knee. Accordingly, recent studies have focused on repair techniques aimed at restoring normal joint contact pressures and kinematics, with promising short- to midterm outcomes reported after root repair.^{6,12,13} However, the duration of follow-up in some of these studies has been limited, making it difficult to draw definitive conclusions regarding long-term efficacy.²¹ In the absence of clearly established treatment algorithms and well-defined patient selection criteria for root repair, nonsurgical treatment has also been proposed as an alternative management strategy. Clinical outcomes after nonsurgical treatment for MMPRT have been inconsistent, with reported conversion rates to total knee arthroplasty (TKA) ranging from 24% to 31%.^{6,31} Consequently, patients with MMPRT are frequently confronted with complex treatment decisions, and shared decision-making represents a critical component of clinical care. Nevertheless, prognostic assessment remains challenging, as outcomes after MMPRT are influenced by multiple factors, including patient characteristics⁵⁰ and radiographic parameters,^{3,29,56} thereby limiting the ability to accurately predict clinical outcomes after treatment.

In recent years, advances in artificial intelligence (AI), including machine learning and deep learning algorithms, have led to its increasing application in orthopaedic research and clinical practice.^{2,24,39,47} These approaches enable the integration of numerous variables into a single predictive framework and have demonstrated considerable potential in predicting survival outcomes,^{20,54} postoperative complications,^{25,30} and patient-reported or clinical assessments.^{22,33,38} Such models may facilitate patient counseling and shared decision-making by supporting more personalized and data-driven treatment planning.^{8,52} However, research focused on the development of clinical prediction models specifically for the treatment of MMPRT remains limited.

This study aimed to develop and internally validate an AI-based prediction model using patient characteristics and clinical data to provide patient-specific risk assessment of clinical failure at 2 and 5 years after nonsurgical treatment of MMPRT. We hypothesized that AI-based prediction models would provide accurate estimates of clinical failure after nonsurgical treatment of MMPRT at both time points, and that the proposed deep learning model would outperform conventional machine learning approaches.

METHODS

Study Patients

This study was approved by our institutional review board. Prospectively collected data were retrospectively reviewed for patients diagnosed with MMPRT between August 2006 and December 2020. MMPRT was defined on magnetic resonance imaging (MRI) as a radial tear located within 9 mm of the posterior tibial attachment of the medial meniscus. Diagnostic MRI findings included the absence of an identifiable meniscus or replacement of the normal low-signal meniscal structure with a high-intensity signal on sagittal images, referred to as the ghost sign; a vertical linear defect at the meniscus root on coronal images; and a radial linear defect at the posterior insertion on axial images.⁵¹

The inclusion criteria were (1) a confirmed diagnosis of MMPRT without associated ligamentous abnormalities on MRI, (2) age <70 years, and (3) nonsurgical treatment with a minimum follow-up duration of 5 years. Patients were excluded if they had Kellgren-Lawrence grade 3 or 4 osteoarthritis, a history of ipsilateral knee surgery, or incomplete clinical data. All patients underwent a structured nonsurgical management program consisting of activity modification with avoidance of heavy load-bearing activities, a minimum 6-week course of supervised physical therapy, and 8 to 12 weeks of anti-inflammatory medication. Patients who demonstrated improvement in pain and functional status after this initial treatment phase were maintained on nonsurgical management and followed annually. Patients who did not improve, or whose symptoms subsequently progressed despite nonsurgical management, proceeded to TKA or osteotomy through a shared decision-making process. These conversions were classified as clinical failure, and no patient was excluded based on their subsequent clinical course.

Based on these criteria, 233 patients (233 knees) were included in the study and constituted the evaluation cohort (Figure 1).

Data Collection

Patient descriptive characteristics and clinical data were obtained from electronic medical records and the institutional database. Baseline imaging included standardized anteroposterior and lateral radiographs; the Rosenberg view, defined as posteroanterior weightbearing radiographs obtained at 45° of knee flexion; and full-length

[‡]Address correspondence to Jong Keun Seon, MD, PhD, Department of Orthopaedic Surgery, Chonnam National University Hwasun Hospital, 322 Seoyang-ro, Hwasun 58128, Republic of Korea (email: seonbell@chonnam.ac.kr).

^{*}Department of Orthopaedic Surgery, Chonnam National University Hwasun Hospital, College of Medicine, Chonnam National University, Hwasun, Republic of Korea.

[†]Department of Electrical Engineering and Computer Science, Daegu Gyeongbuk Institute of Science and Technology (DGIST), Daegu, Republic of Korea.

H.Y.Y. and I.B. contributed equally to this work.

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Ethical approval for this study was obtained from Chonnam National University Hwasun Hospital (IRB No. CNUHH-2025-185).

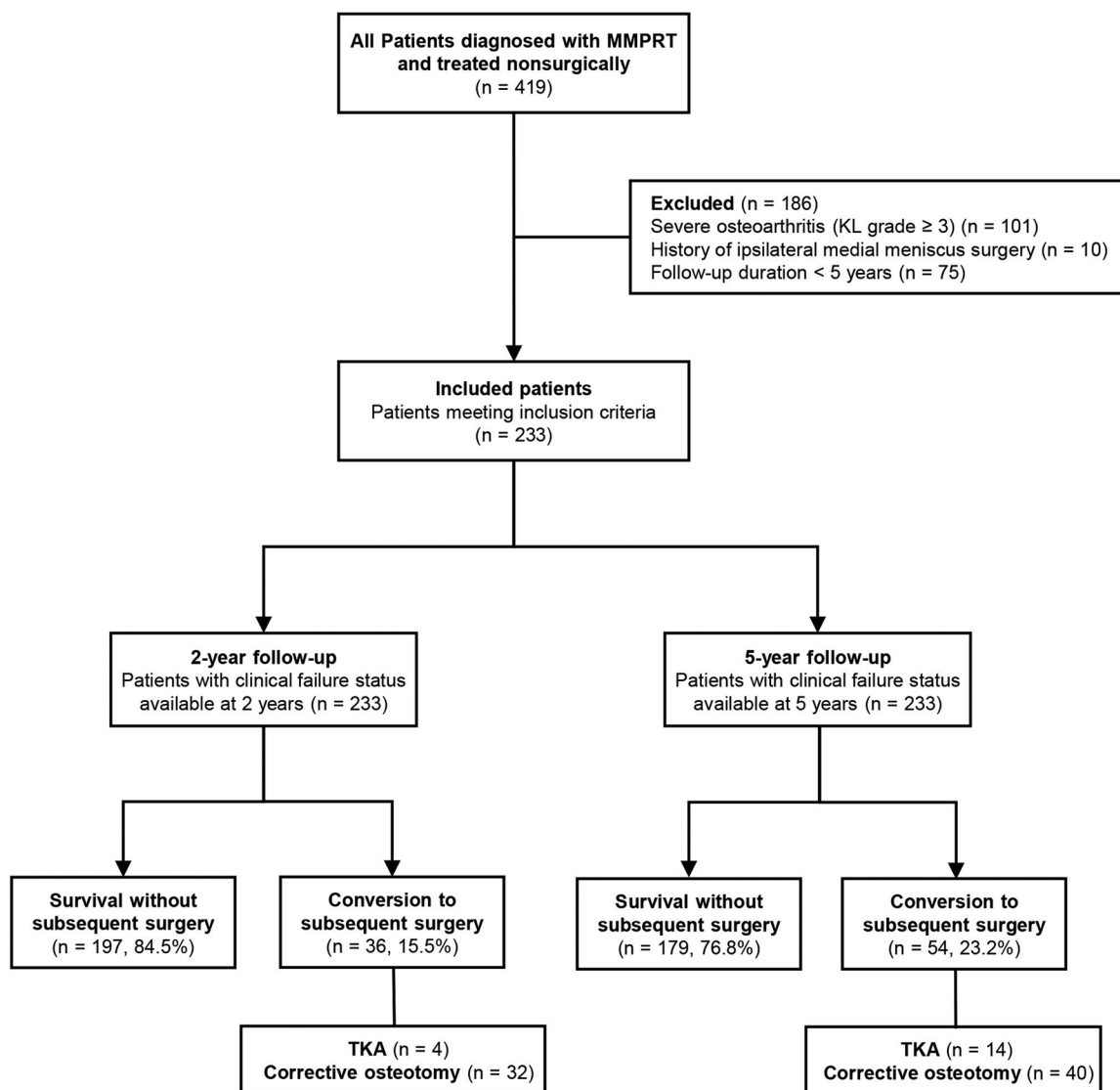


Figure 1. Flowchart illustrating patient selection and the analytical workflow. KL, Kellgren-Lawrence; MMPRT, medial meniscus posterior root tear; TKA, total knee arthroplasty.

standing weightbearing radiographs (FLWBRs). Radiographic assessments included measurement of the hip-knee-ankle (HKA) angle, medial proximal tibial angle (MPTA), lateral distal femoral angle (LDFA), posterior tibial slope (PTS), joint line convergence angle, medial joint space width, and Kellgren-Lawrence grade. Medial joint space width was measured from the center of the medial femoral condyle to the center of the medial tibial plateau on both the Rosenberg view and FLWBRs.⁵⁸ Baseline MRI scans were reviewed to assess medial and lateral tibial slope,²⁷ medial meniscus extrusion (MME),¹⁷ the presence of the posterior shiny-corner lesion (PSCL),¹⁰ bone marrow edema (BME),¹⁸ subchondral insufficiency fracture of the knee (SIFK) involving the medial femoral condyle,⁵⁷ MRI Osteoarthritis Knee Score (MOAKS) effusion

grade,²⁸ and cartilage lesions. Medial and lateral PTS were measured on sagittal MRI scans at the center of the tibial plateau using the method described by Hudek et al.²⁷ MME was quantified in millimeters as the distance from the medial tibial plateau margin to the outer border of the medial meniscus on the coronal image at the mid-point of the medial femoral condyle, according to the method described by Costa et al.¹⁷ A relative MME value was additionally calculated as the ratio of absolute MME to the total width of the medial meniscus. The PSCL was defined as a focal peripheral hyperintense lesion at the posterior aspect of the medial tibial plateau corresponding to meniscus-covered regions, as previously described.¹⁰ BME was defined as a focal alteration in bone marrow signal.¹⁸ SIFK was defined as a low-signal-intensity line

beneath the subchondral bone on T1-weighted images.⁵⁷ Initial cartilage arthrosis in the medial compartment was graded according to the modified Outerbridge classification system.⁴⁹ Radiographic and MRI assessments were performed by 2 independent investigators (G.B. and Y.Z.). Each observer repeated all measurements twice, with a 4-week interval between assessments, to evaluate intra-observer and interobserver reliability. Intraclass correlation coefficients (ICCs) were calculated for each parameter, with all values demonstrating good agreement, defined as an ICC >0.80.

Primary Study Outcome

The primary outcome of the study was conversion to subsequent surgery, defined as TKA or corrective osteotomy at 2 and 5 years after nonsurgical treatment. Conversion to surgery was classified as clinical failure, whereas the absence of subsequent surgical intervention was defined as survival free from surgery under nonsurgical management. The occurrence and timing of subsequent surgical procedures were determined through comprehensive review of medical records and follow-up documentation for all patients. The 2- and 5-year time points were selected because progression of osteoarthritis and conversion to TKA have been reported to be particularly frequent 5 years after nonsurgical treatment for MMPRT, and because reliable follow-up data were available at these intervals.

Data Preprocessing and Feature Engineering

Through the data collection process, a maximum of 24 features per patient were obtained from electronic medical records and radiological assessments and were eligible for the development of the prediction model. These features underwent a preprocessing stage before model training and development. First, missing data, which occurred for various reasons, were explicitly marked to allow the model to recognize missingness as an informative feature. Second, normalization was applied to ensure that the mean and variance of each feature across all patients were statistically standardized. Third, nonnumeric categorical variables were encoded into integer values according to their respective classes.

Subsequently, data augmentation strategies were applied to improve robustness during the training phase of the AI model. Noise injection was introduced into the raw data to reduce overfitting to specific numeric patterns. In addition, individual features or groups of features within the same domain were randomly masked simultaneously, thereby encouraging the model to extract meaningful information from all available inputs rather than relying disproportionately on easily discriminable features.

AI Model Development

The preprocessed data set was used to train 6 AI models, including 5 conventional machine learning models—Elastic

Net logistic regression (ElasticNetLR), multilayer perceptron, support vector machine, random forest, and Extreme Gradient Boosting (XGBoost)—as well as one proposed deep learning model, the Grouped Graph Attention (GGAT) network.

The architecture of the proposed GGAT model was specifically designed to enable accurate outcome prediction through structured feature integration (Figure 2). A group-based hierarchical feature processing architecture was implemented, as previously described.^{4,62} Input variables were initially organized into 4 feature domains consisting of patient characteristics, clinical presentation, radiographic parameters, and MRI findings. Within each domain, a neural network layer aggregated the input variables into 4 latent features through intragroup summarization. These latent features were subsequently refined through intergroup information aggregation across domains. A graph attention network was incorporated to assign differential importance to domain-level features and to enhance information flow based on learned relationships among domains. Finally, a neural network classifier was applied to estimate the probability of treatment failure and survival.

Feature Selection Using Shapley Additive exPlanations Analysis and Recursive Feature Elimination

The proposed deep learning model was further utilized during the feature selection phase to assess the relative importance of individual input variables. To quantify the contribution and direction of each predictor to the model output, Shapley Additive exPlanations (SHAP) analysis was applied.⁴³ Through the use of SHAP, the predicted probability generated by the model trained with all input variables was decomposed into individual feature contributions, allowing the magnitude and direction of each variable's influence on the prediction to be quantified. Subsequently, features with the lowest relative importance were removed using the Recursive Feature Elimination (RFE) method. Feature selection was performed by iteratively alternating between SHAP analysis and RFE to identify the minimal subset of the most informative features that optimized model performance for both 2-year and 5-year outcomes. This process resulted in selection of the following features: baseline mechanical HKA angle, symptom duration, body mass index (BMI), age, BME, baseline LDFA, SIFK, initial cartilage lesion, MOAKS effusion grade, and baseline MPTA.

Model Training and Evaluation

All AI models were configured as binary classifiers, with up to 24 input features used to estimate the probability of treatment survival or failure. For model development, the 2-year and 5-year data sets were randomly divided into training and test sets in an 80% to 20% ratio. The 6 AI models were first validated on the training set using a standard cross-validation framework. A stratified 5-fold

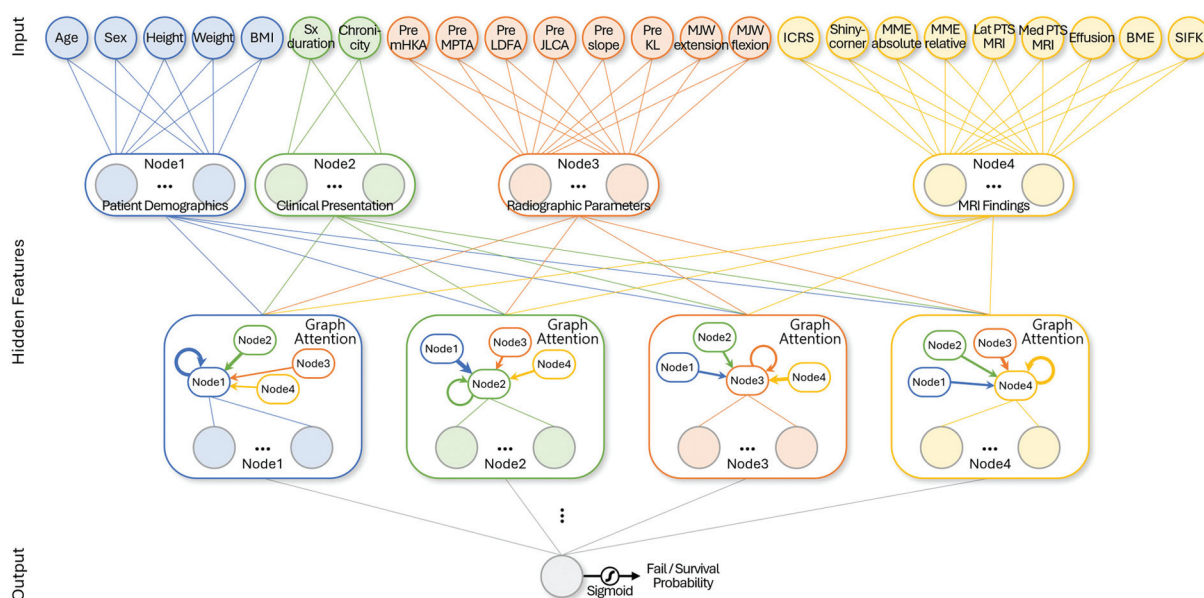


Figure 2. Architecture of the Grouped Graph Attention (GGAT) model for predicting outcomes after nonsurgical treatment of medial meniscus posterior root tear (MMPRT). Input variables were grouped into 4 feature domains: patient characteristics (blue), clinical presentation (green), radiographic parameters (orange), and magnetic resonance imaging (MRI) findings (yellow). Connections between group nodes were weighted by attention coefficients, with stronger relationships assigned higher values and weaker relationships lower values. Aggregated outputs were used to predict survival versus subsequent surgical intervention. BME, bone marrow edema; ICRS, International Cartilage Regeneration & Joint Preservation Society (cartilage lesion grade, assessed using the modified Outerbridge classification); JLCA, joint line convergence angle; KL, Kellgren-Lawrence; LDFA, lateral distal femoral angle; mHKA, mechanical hip-knee-ankle angle; MJW, medial joint space width; MME, medial meniscus extrusion; MPTA, medial proximal tibial angle; SIFK, subchondral insufficiency fracture of the knee; Sx, symptom.

cross-validation strategy with 3 iterations was used to preserve outcome distribution and to reduce training variance, thereby ensuring stable model optimization. After training, the fully optimized models were evaluated using the independent test set. Model performance was assessed using 7 predefined metrics, including accuracy, precision, sensitivity, specificity, F1 score, Brier score, and area under the receiver operating characteristic curve (AUC). Cross-validation results are reported as the mean metric value across all iterations with corresponding 95% confidence interval, whereas test set performance is reported using metrics derived from evaluation of the final trained models. Because this retrospective prediction model study was based on a fixed cohort, no formal a priori power calculation was performed. Model adequacy was assessed based on the number of available outcome events relative to the final predictor set and the stability of internal validation procedures.

Statistical Analysis

Descriptive statistics were used to characterize the study population. Continuous variables are presented as mean with standard deviation, and categorical variables are reported as count with percentage. Data preprocessing and statistical analyses were performed using Python Version 3.10 (Python Software Foundation). Statistical

significance was defined as a 2-sided P value $<.05$ for all analyses. Data preprocessing, feature engineering, model development, and model training and evaluation were performed by an investigator (I.B.) with expertise in AI and machine learning.

Web Application

The final deep learning model, which demonstrated the highest performance metrics after feature selection, was deployed as a freely accessible online prediction tool (<https://ihbae.com/ai-mmprt>). This web-based application performs all computations entirely on device, ensuring that entered patient information is processed locally and remains private. The current version of the model is intended for informational purposes only. It remains in the validation phase to assess its predictive performance, and further external validation is required to confirm robustness before clinical implementation.

RESULTS

A total of 233 patients with a mean age of 59.7 ± 6.6 years were diagnosed with MMPRT and followed for a mean duration of 7.0 ± 3.9 years after diagnosis. Baseline descriptive characteristics and clinical data are

TABLE 1
Baseline Characteristics and Clinical Data of the Study Cohort^a

	Value
Age, y	59.7 ± 6.6
Sex	
Female	211 (90.6)
Male	22 (9.4)
Body mass index, kg/m ²	25.4 ± 3.1
Symptom duration, mo	9.7 ± 10.0
Injury chronicity ^b	105 (45.1)
Radiographic findings	
Kellgren-Lawrence grade 0/1/2	34/92/107
Mechanical HKA angle, deg ^c	4.1 ± 2.9
MPTA, deg	86.2 ± 2.1
LDFA, deg	88.6 ± 2.2
JLCA, deg	1.8 ± 1.4
PTS, deg	8.4 ± 3.2
Width of medial joint space using the FLWBR, mm	3.9 ± 0.6
Width of medial joint space using the Rosenberg view, mm	3.5 ± 0.7
MRI findings	
Medial PTS, deg	5.8 ± 1.8
Lateral PTS, deg	6.3 ± 1.6
Absolute MME, mm	3.7 ± 0.9
Relative MME, %	29.5 ± 6.3
Posterior shiny-corner lesion	89 (38.2)
BME	121 (51.9)
SIFK	94 (40.3)
MOAKS effusion grade 0/1/2/3	39/96/72/26
Cartilage arthrosis grade 0/1/2/3/4 ^d	54/6/100/72/1
Duration of follow-up, y	7.0 ± 3.9
Conversion to subsequent surgery (at any follow-up time point)	
TKA	29 (12.4)
Corrective osteotomy	46 (19.7)

^aValues are presented as mean ± SD or n (%). BME, bone marrow edema; FLWBR, full-length standing weightbearing radiograph; HKA, hip-knee-ankle; JLCA, joint line convergence angle; LDFA, lateral distal femoral angle; MME, medial meniscus extrusion; MOAKS, MRI Osteoarthritis Knee Score; MPTA, medial proximal tibial angle; MRI, magnetic resonance imaging; PTS, posterior tibial slope; SIFK, subchondral insufficiency fracture of the knee; TKA, total knee arthroplasty.

^bInjury chronicity was categorized as chronic if the interval between injury or symptom onset and imaging was ≥12 weeks.

^cA positive angle indicates varus alignment, whereas a negative angle indicates valgus alignment.

^dCartilage arthrosis was graded according to the modified Outerbridge classification system.

summarized in Table 1. Conversion to subsequent surgical treatment was observed in 15.5% of patients (36/233) at a minimum follow-up of 2 years and in 23.2% of patients (54/233) at a minimum follow-up of 5 years. Among patients who underwent surgical conversion, the mean interval from diagnosis to surgery was 39.8 ± 38.0 months.

Quantitative Performance of AI Models

The cross-validation performance for the prediction of treatment prognosis at 2 and 5 years is presented in Table 2. For the 2-year prediction, ElasticNetLR demonstrated the best performance among the conventional machine learning models, with an accuracy of 0.85, an AUC of 0.69, and a Brier score of 0.12. Across the conventional machine learning models, performance generally declined for the 5-year prediction, likely reflecting the increased complexity associated with midterm prognostication.

Accordingly, for the 5-year outcome, the comparatively simple Elastic Net model showed reduced performance, whereas the more advanced XGBoost model achieved the best results among the machine learning approaches, with an accuracy of 0.79, an AUC of 0.80, and a Brier score of 0.15. In contrast, the proposed deep learning model, the GGAT network, demonstrated stable performance across both the 2- and 5-year follow-up periods and significantly outperformed the conventional machine learning models. The GGAT model achieved the highest overall predictive performance, with accuracy ranging from 0.87 to 0.89, an AUC of 0.87 at both time points, and a Brier score ranging from 0.09 to 0.11.

The performance of the final AI models on the independent test set is presented in Table 3. Consistent with the cross-validation findings, the GGAT model demonstrated more robust performance than the conventional machine learning models at both 2 and 5 years.

TABLE 2
Cross-Validation Performance of Models for Predicting 2-Year and 5-Year
Surgical Conversion After MMPRT^a

	2-y Prediction						5-y Prediction					
	ElasticNetLR	MLP	SVM	RF	XGBoost	Proposed GGAT	ElasticNetLR	MLP	SVM	RF	XGBoost	Proposed GGAT
Accuracy	0.85 (0.84 to 0.86)	0.84 (0.83 to 0.85)	0.84 (0.83 to 0.85)	0.83 (0.80 to 0.86)	0.83 (0.81 to 0.84)	0.89 (0.87 to 0.91)	0.74 (0.68 to 0.80)	0.77 (0.76 to 0.78)	0.77 (0.76 to 0.78)	0.75 (0.72 to 0.77)	0.79 (0.77 to 0.81)	0.87 (0.85 to 0.89)
Precision	0.85 (0.84 to 0.86)	0.84 (0.83 to 0.85)	0.84 (0.83 to 0.85)	0.86 (0.84 to 0.87)	0.85 (0.84 to 0.86)	0.90 (0.87 to 0.92)	0.78 (0.77 to 0.79)	0.77 (0.76 to 0.78)	0.77 (0.76 to 0.78)	0.78 (0.77 to 0.80)	0.81 (0.80 to 0.83)	0.87 (0.85 to 0.89)
Sensitivity	1.00 (1.00 to 1.00)	1.00 (1.00 to 1.00)	1.00 (1.00 to 1.00)	0.96 (0.94 to 0.99)	0.97 (0.95 to 0.99)	0.99 (0.98 to 1.00)	0.94 (0.83-1.00)	1.00 (1.00 to 1.00)	1.00 (1.00 to 1.00)	0.93 (0.90 to 0.96)	0.94 (0.92 to 0.97)	0.98 (0.96 to 0.99)
Specificity	0.02 (-0.01 to 0.05)	0.00 (0.00 to 0.00)	0.00 (0.00 to 0.00)	0.13 (0.06 to 0.20)	0.06 (0.01 to 0.10)	0.37 (0.25 to 0.50)	0.10 (-0.02 to 0.21)	0.00 (0.00 to 0.00)	0.00 (0.00 to 0.00)	0.15 (0.08 to 0.21)	0.28 (0.19 to 0.37)	0.50 (0.40 to 0.60)
F1	0.92 (0.91 to 0.92)	0.92 (0.91 to 0.92)	0.92 (0.91 to 0.92)	0.91 (0.89 to 0.92)	0.90 (0.89 to 0.91)	0.94 (0.93 to 0.95)	0.83 (0.75 to 0.91)	0.87 (0.87 to 0.87)	0.87 (0.87 to 0.87)	0.85 (0.83 to 0.87)	0.87 (0.86 to 0.89)	0.92 (0.91 to 0.93)
Brier score	0.12 (0.12 to 0.13)	0.16 (0.15 to 0.17)	0.13 (0.12 to 0.13)	0.18 (0.15 to 0.21)	0.13 (0.12 to 0.14)	0.09 (0.07 to 0.10)	0.19 (0.16 to 0.22)	0.23 (0.22 to 0.24)	0.17 (0.16 to 0.17)	0.23 (0.20 to 0.27)	0.15 (0.13 to 0.16)	0.11 (0.09 to 0.12)
AUC	0.69 (0.65 to 0.74)	0.48 (0.44 to 0.53)	0.70 (0.65 to 0.75)	0.59 (0.53 to 0.65)	0.67 (0.61 to 0.72)	0.87 (0.82 to 0.92)	0.63 (0.59 to 0.67)	0.41 (0.38 to 0.44)	0.67 (0.62 to 0.71)	0.63 (0.57 to 0.69)	0.80 (0.76 to 0.84)	0.87 (0.84 to 0.91)

^aValues are presented as mean (95% CI). AUC, area under the receiver operating characteristic curve; ElasticNetLR, Elastic Net logistic regression; GGAT, Grouped Graph Attention; MLP, multilayer perceptron; MMPRT, medial meniscus posterior root tear; RF, random forest; SVM, support vector machine; XGBoost, Extreme Gradient Boosting.

TABLE 3
Test Set Evaluation Results for Predicting Subsequent Surgery After MMPRT
at 2-Year and 5-Year Follow-up^a

	2-y Prediction						5-y Prediction					
	ElasticNetLR	MLP	SVM	RF	XGBoost	Proposed GGAT	ElasticNetLR	MLP	SVM	RF	XGBoost	Proposed GGAT
Accuracy	0.85	0.85	0.85	0.85	0.85	0.91	0.79	0.77	0.77	0.77	0.81	0.83
Precision	0.85	0.85	0.85	0.87	0.85	0.91	0.78	0.77	0.77	0.79	0.81	0.87
Sensitivity	1.00	1.00	1.00	0.98	1.00	1.00	1.00	1.00	1.00	0.94	0.97	0.92
Specificity	0.00	0.00	0.00	0.14	0.00	0.43	0.09	0.00	0.00	0.18	0.27	0.55
F1	0.92	0.92	0.92	0.92	0.92	0.95	0.88	0.87	0.87	0.86	0.89	0.89
Brier score	0.12	0.15	0.11	0.20	0.12	0.09	0.16	0.23	0.16	0.27	0.18	0.16
AUC	0.54	0.41	0.70	0.46	0.63	0.77	0.71	0.54	0.66	0.51	0.61	0.69

^aAUC, area under the receiver operating characteristic curve; ElasticNetLR, Elastic Net logistic regression; GGAT, Grouped Graph Attention; MLP, multilayer perceptron; MMPRT, medial meniscus posterior root tear; RF, random forest; SVM, support vector machine; XGBoost, Extreme Gradient Boosting.

Feature Importance of Predictive Models

The SHAP summary plot and mean absolute SHAP values are presented in Figure 3 to illustrate the relative contribution of individual features to predictions generated by the GGAT model. The most influential predictors, ranked in descending order of mean absolute SHAP value, were baseline mechanical HKA angle, symptom duration, BMI, age, BME, baseline LDFA, SIFK, initial cartilage lesion, MOAKS effusion grade, and baseline MPTA. The SHAP summary plot further indicated that age and baseline MPTA were positively associated with the predicted probability of survival, whereas the remaining features exhibited negative contributions. RFE analysis confirmed that this reduced set of variables preserved model performance, as sequential removal of less influential features did not reduce predictive accuracy.

Exploratory SHAP dependence analysis showed that the contribution of symptom duration to predicted survival progressively decreased and transitioned from positive to negative at approximately 14.9 months (Figure 4).

Patient-Level Feature Importance

Representative SHAP plots at the individual patient level are presented in Figure 5. These examples illustrate that the relative influence of specific features on survival prediction varied between the 2-year and 5-year follow-up periods.

Web Application

The pretrained deep learning model incorporating the 10 selected features was integrated into a freely accessible online prediction tool. On entry of the 10 patient-specific

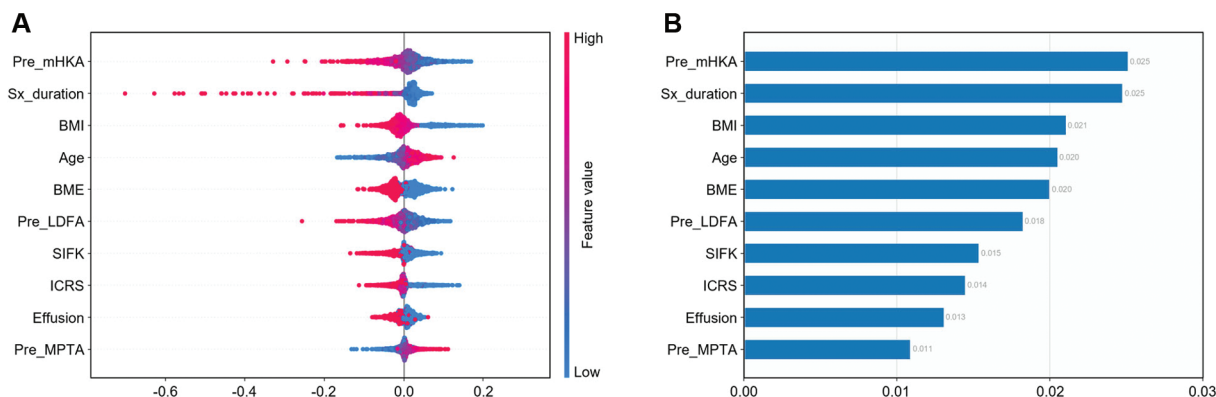


Figure 3. Feature selection using Shapley Additive exPlanations (SHAP) analysis and Recursive Feature Elimination (RFE). (A) The SHAP summary plot illustrates the influence of individual data points on survival prediction across input variables. Color represents the feature value, with red indicating higher values and blue indicating lower values. The horizontal position reflects the direction and magnitude of each feature's contribution to the model prediction, with values to the right increasing and values to the left decreasing the predicted probability of survival. (B) Mean absolute SHAP values summarize the relative importance of variables in descending order. Sequential feature elimination using RFE resulted in a final set of 10 predictors. BME, bone marrow edema; BMI, body mass index; ICRS, International Cartilage Regeneration & Joint Preservation Society (cartilage lesion grade, assessed using the modified Outerbridge classification); LDFA, lateral distal femoral angle; mHKA, mechanical hip-knee-ankle angle; MPTA, medial proximal tibial angle; SIFK, subchondral insufficiency fracture of the knee; Sx, symptom.

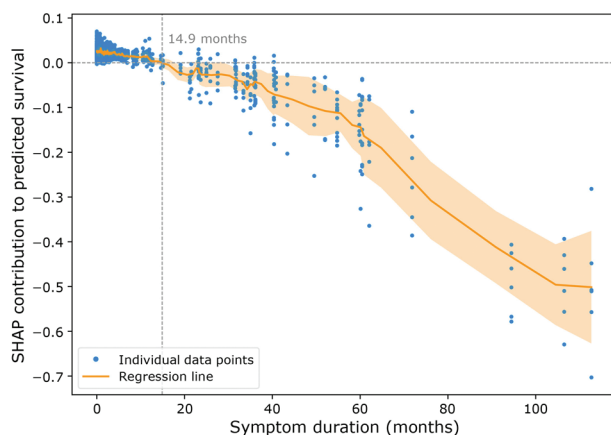


Figure 4. Shapley Additive exPlanations (SHAP) dependence plot for symptom duration. Each point represents an individual patient, and the orange line represents the fitted regression line. The contribution of symptom duration to predicted survival progressively decreased with increasing symptom duration and transitioned from positive to negative at approximately 14.9 months, representing a model-derived transition point rather than a definitive clinical cutoff.

input features, the tool generates predicted treatment outcomes along with the corresponding probabilities at 2 and 5 years. In addition, the application provides a survival probability simulation graph that illustrates changes in the predicted 5-year treatment success probability in response to variations in BMI, thereby offering supportive information for clinical guidance.

DISCUSSION

The principal finding of this study was that a deep learning-based prediction model, the GGAT network, demonstrated accurate and reliable performance in identifying patients at risk of clinical failure at both 2 and 5 years after nonsurgical treatment of MMPRT. By leveraging routinely available patient descriptive and clinical variables, the model enabled patient-specific risk assessment, with potential utility in supporting clinical decision-making and patient counseling regarding nonsurgical management strategies. These findings suggest that integration of deep learning-based tools into clinical workflows may help optimize patient management and facilitate more personalized treatment planning.

Given the limited evidence guiding management of MMPRT and the adverse effects of these injuries on knee kinematics and long-term joint health, deep learning-based approaches may enhance understanding of MMPRT pathology and improve the precision of prognostic assessment, with the potential to uncover novel disease-related insights.^{5,35,55} Beyond improving prognostic accuracy, the present findings offer clinically relevant insights into factors associated with failure after nonsurgical treatment of MMPRT. In particular, BMI emerged as an important predictor, underscoring the value of individualized risk estimation in informing patient counseling regarding modifiable factors associated with treatment outcomes (Figure 6). By integrating risk estimates derived from readily available clinical variables, this approach may support more informed and individualized discussions between clinicians and patients when considering nonsurgical treatment options.

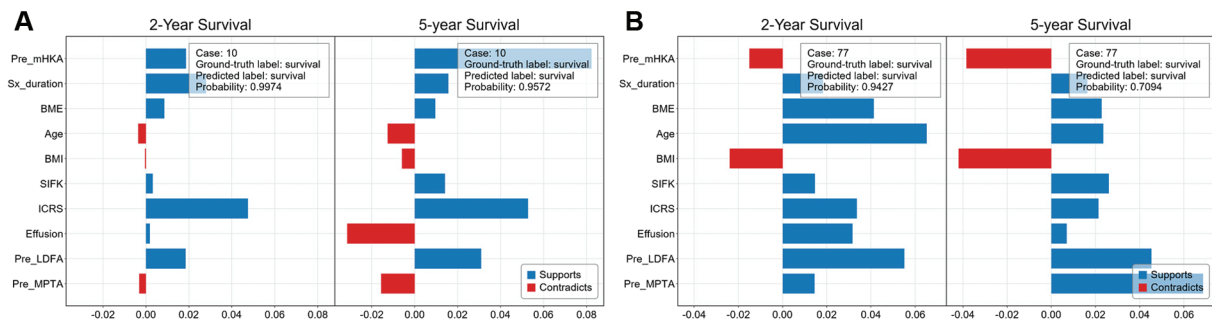


Figure 5. Case-based graphical illustration demonstrating clinical application of the developed prediction model. Selected features were used to compute individualized probabilities of survival at 2-year and 5-year follow-up. Variables highlighted in blue indicate factors supporting survival, whereas variables highlighted in red indicate factors opposing survival. Two representative cases are shown. (A) Case 10 demonstrated survival probabilities of 99.7% at 2 years and 95.7% at 5 years, consistent with stable survival over time. (B) Case 77 demonstrated a decline in survival probability from 94.3% at 2 years to 70.9% at 5 years as opposing features became increasingly influential. BME, bone marrow edema; BMI, body mass index; ICRS, International Cartilage Regeneration & Joint Preservation Society (cartilage lesion grade, assessed using the modified Outerbridge classification); LDFA, lateral distal femoral angle; mHKA, mechanical hip-knee-ankle angle; MPTA, medial proximal tibial angle; SIFK, subchondral insufficiency fracture of the knee; Sx, symptom.

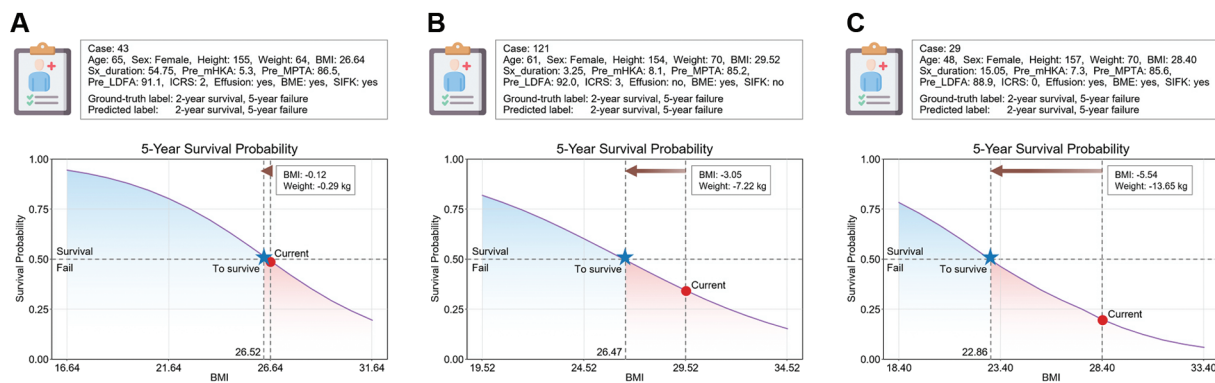


Figure 6. Model-based clinical suggestions for improving 5-year survival according to body mass index (BMI). Three representative patients (cases 43, 121, and 29) who demonstrated survival at 2 years but experienced clinical failure at 5 years are shown. Using all input variables, the model estimated changes in the predicted probability of 5-year survival as BMI varied. (A) In case 43, a modest reduction in BMI was associated with a shift toward a higher predicted probability of 5-year survival. In contrast, (B) case 121 and (C) case 29 required larger reductions in BMI to achieve comparable increases in predicted survival probability. BME, bone marrow edema; ICRS, International Cartilage Regeneration & Joint Preservation Society (cartilage lesion grade, assessed using the modified Outerbridge classification); LDFA, lateral distal femoral angle; mHKA, mechanical hip-knee-ankle angle; MPTA, medial proximal tibial angle; SIFK, subchondral insufficiency fracture of the knee; Sx, symptom.

At present, no clear consensus exists regarding the optimal treatment strategy for MMPRT, and clinical outcomes after nonsurgical management remain variable. Historically, MMPRTs were treated either nonoperatively or with partial meniscectomy; however, recent trends have increasingly favored meniscal preservation through root repair.⁷ Krych et al³⁷ reported that nonsurgical treatment of MMPRT was associated with poor clinical outcomes, a relatively high rate of conversion to TKA, and progression of osteoarthritis at the 5-year follow-up, particularly among patients with higher baseline Kellgren-Lawrence grades. In a subsequent study, Krych et al³⁶ found that progression to arthroplasty occurred in 54% of patients

treated with partial meniscectomy and in 34.6% of those managed nonoperatively at mean follow-up durations of 54.3 and 30.2 months, respectively. Root repair has been performed with increasing frequency in light of accumulating clinical evidence demonstrating unfavorable long-term outcomes after nonsurgical management.^{36,37,40,46,59} Ahn et al¹ compared patients treated with root repair with those managed nonoperatively and reported superior clinical outcomes in the repair group. Given the critical role of the meniscus root in maintaining hoop stress and preventing meniscal extrusion during compressive loading, a primary rationale for root repair has been to mitigate osteoarthritis progression and reduce the risk of

conversion to TKA. In a comparative study, Dragoo et al¹⁹ demonstrated substantially lower rates of conversion to TKA and improved functional outcomes in patients undergoing root repair compared with those treated nonoperatively. Similarly, a systematic review comparing root repair with nonrepair strategies, including meniscectomy and nonsurgical treatment, reported a lower rate of radiographic osteoarthritis progression in the repair group. Specifically, Kellgren-Lawrence grade progression of at least one grade was observed in 25% (29/114) of patients undergoing repair at a mean follow-up of 45.4 months, compared with 40% (38/96) of patients treated without repair at a mean follow-up of 62.7 months.⁴¹ Krivich et al³⁴ further reported that 9.8% (8/82) of patients undergoing MMPRT repair progressed to TKA at mid- to long-term follow-up, whereas conversion occurred in 36% (22/61) of patients treated with meniscectomy. In the present study, the rate of conversion to TKA after nonsurgical treatment of MMPRT was 12.4% (29/233) at a mean follow-up of 7.0 years. This rate falls between those reported after MMPRT repair and partial meniscectomy in prior studies. Nevertheless, prospective investigations with longer-term follow-up are required to more definitively characterize treatment-specific differences in clinical failure rates across management strategies for MMPRT.

In the present study, a deep learning-based prediction model demonstrated accurate and reliable performance in estimating the risk of clinical failure at both 2 and 5 years after nonsurgical treatment of MMPRT. In the clinical setting, individualized risk estimates of this nature may facilitate shared decision-making by helping patients develop more informed expectations regarding treatment outcomes. Although multiple predictors contribute to the model, their combined effects are integrated into a single patient-specific risk estimate through the accompanying web-based tool. In addition, patient-level SHAP visualizations may help clinicians identify the principal factors contributing to each prediction and facilitate patient counseling. The identified predictors, including baseline mechanical HKA angle, symptom duration, BMI, age, BME, LDFA, SIFK, cartilage status, effusion grade, and MPTA, are consistent with factors previously reported to influence outcomes in patients with MMPRT.⁸ Varus alignment has been associated with increased medial tibiofemoral contact pressures and elevated stress on the posterior medial meniscus.^{1,61} Several clinical studies have identified varus alignment as a risk factor for MMPRT,^{9,45} as well as for inferior clinical outcomes and higher failure rates after treatment.^{14,45} Our findings suggest that the prognostic influence of symptom duration may not be adequately captured by the conventional 12-week definition of chronicity, supporting the use of symptom duration as a continuous variable. Obesity contributes to increased mechanical loading on articular cartilage, resulting in tissue damage, and adipose tissue releases adipokines that promote cartilage inflammation

and degradation.⁵³ Higher BMI has been associated with greater osteoarthritis progression in patients with MMPRT.⁴⁸ Consistent with our findings, Helito et al²⁶ identified higher BMI as an independent predictor of failure after root repair, suggesting that elevated BMI may adversely affect outcomes across different treatment strategies. Zanetti et al⁶³ reported that 40.6% of patients with SIFK were overweight or obese, suggesting a complex interrelationship between metabolic factors and subchondral bone pathology. SIFK has been frequently associated with concomitant cartilage loss and meniscal injury and has been linked to progressive osteoarthritis and subsequent conversion to TKA.^{57,60} It is therefore reasonable to propose an association among MMPRT, joint space narrowing, and SIFK. Loss of meniscus root function in MMPRT disrupts hoop stress transmission and increases tibiofemoral contact pressure, which may contribute to joint space narrowing and progressive cartilage degeneration.^{5,16,44} These biomechanical alterations may further concentrate stress on the subchondral bone, thereby contributing to the development of SIFK.⁴⁸ In the present cohort, the mean MPTA and LDFA values were $86.2^\circ \pm 2.1^\circ$ and $88.6^\circ \pm 2.2^\circ$, respectively, which are consistent with previously reported epidemiological characteristics of MMPRT.^{5,11,32} Taken together, deformity and bone morphology of the distal femur and proximal tibia were associated with clinical failure, underscoring the importance of comprehensive lower limb alignment assessment when evaluating treatment strategies for MMPRT. Given the prognostic relevance of these clinical features, early identification of risk factors may support prompt risk stratification and inform consideration of treatment strategies, including MMPRT repair with or without corrective osteotomy. In addition, this information may assist clinicians in discussing treatment options and expected outcomes with patients in a more individualized and evidence-based manner.

Limitations

This study has several limitations that warrant consideration. First, the retrospective and nonrandomized study design introduces the potential for selection bias related to patient inclusion. Second, follow-up was limited to the midterm, with conversion to subsequent surgical procedures evaluated at 2-year and 5-year endpoints. Although prior studies have demonstrated that progression of osteoarthritis and conversion to TKA most frequently occur within this time frame, longer-term outcomes were not assessed. Given the progressive nature of osteoarthritis, future studies incorporating extended follow-up may provide additional insight into long-term prognosis after nonsurgical treatment of MMPRT. Third, although a comprehensive set of patient descriptive characteristics and clinical variables was included, other potentially relevant factors may not have been captured and could have influenced model performance. Fourth, the predominance of middle-aged female patients in the study cohort may limit the generalizability of these findings to broader

[§]References 1, 5, 9, 11, 14, 16, 26, 32, 44, 45, 48, 57, 60, 61, 63.

patient populations. Finally, the number of clinical failure events was relatively limited, which may have affected the precision of the performance estimates and the generalizability of the model. Although repeated stratified cross-validation and independent test set evaluation supported the internal stability of the model, external validation in larger and more diverse multicenter cohorts remains necessary to confirm its robustness and generalizability. Despite these limitations, the findings may provide clinically relevant information to support clinician-patient discussions regarding nonsurgical management of MMPRT and may serve as a foundation for future externally validated studies.

CONCLUSION

The deep learning-based GGAT network demonstrated accurate prediction of clinical failure at both 2 and 5 years after nonsurgical treatment of MMPRT. These findings underscore the potential value of deep learning-based risk assessment tools in supporting clinical decision-making and patient counseling. Such approaches may contribute to optimization of patient management and facilitate more personalized treatment planning.

ORCID iDs

Hong Yeol Yang  <https://orcid.org/0000-0001-8730-9040>

Inhwan Bae  <https://orcid.org/0000-0003-1884-2268>

Ji Won Kim  <https://orcid.org/0000-0003-3084-9200>

Jong Keun Seon  <https://orcid.org/0000-0002-6450-2339>

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